Coherence time in biological oscillator assemblies bounds the rate of state registration

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Abstract

Cognitive and biological processing depends on coherent oscillations across distributed assemblies. Yet the physical principles determining how fast these assemblies can synchronize—and thus how fast neural processes associated with cognition can proceed—remain incompletely formalized. We derive a quantitative framework showing that coherence time in coupled oscillator networks scales exponentially with coordination depth. For M semi-independent modules requiring phase alignment within tolerance ε at coherence r, the minimum interval between irreversible state registrations ("commits") is dominated by rare-event first-passage time into low-measure alignment sets on the configuration manifold. This produces a fundamental speed-flexibility trade-off: increasing coordination depth M expands combinatorial flexibility but slows commits exponentially; increasing coherence r accelerates synchronization but restricts dynamics to low-dimensional attractors. We situate this coordination limit within a hierarchy of physical bounds—quantum speed limits, Landauer-Margolus-Levitin thermodynamic constraints, and energy-time uncertainty relations—showing that in biological neural systems, these quantum and thermodynamic bounds are typically non-binding; the dominant constraint is the time required for distributed oscillatory assemblies to achieve sufficient phase coherence. While neural oscillators provide the most empirically accessible measurements, the framework applies to any biological system whose function requires coordinated phase relations and sparse commitment events, from chromosomal segregation to circadian networks. We provide parameter estimation protocols and testable predictions including alpha entrainment effects on temporal acuity and dual-loop dissociations under arousal.

Keywords: biological oscillations, coherence time, coordination depth, temporal resolution, quantum speed limits, Landauer bound, coupled oscillators

1 Introduction

1.1 Processing speed in biological oscillator assemblies

A fundamental insight from systems neuroscience is that cognition emerges from coherent oscillations across neural assemblies, not merely from individual spike rates. Working memory maintenance requires sustained theta-gamma phase-amplitude coupling [1, 6]. Attention selectively enhances inter-areal synchronization in gamma band [4, 7]. Perceptual binding depends on transient phase alignment across sensory cortices [5, 8]. Long-range communication occurs preferentially during coherent states [9]. The "communication through coherence" framework [10] has become central to understanding neural computation.

Yet despite extensive empirical characterization, the physical principles governing *how fast* distributed assemblies can achieve coherence—and thus how quickly the neural processes underlying cognitive operations can proceed—remain incompletely formalized. Why does perceptual binding require 30–50 ms rather than 3 ms or 300 ms? Why do larger assemblies integrating more information process more slowly? What determines the relationship between oscillation frequency and temporal acuity?

1.2 Coherence time as a physical bound on processing rate

Previous work established that living intelligence is an emergent property of high-dimensional, self-sustaining dynamical systems that maintain coherence against decoherence [11]. The present manuscript addresses the complementary question: given such a coherent high-dimensional system, what sets the fastest timescale on which it can register change and update its state?

We formalize this as the minimum interval between irreversible "commit" events—transitions in which distributed dynamics become readable to downstream mechanisms. This commit interval is bounded by a coordination term (coherence time as a first-passage problem on a product manifold) together with quantum, noise, and power limits.

Operational definition of commits in neural systems. A commit is operationally detected when distributed oscillatory dynamics cross a threshold that enables downstream readout. Empirical signatures include: (i) phase-locking index exceeding threshold across specified modules for a minimum duration, (ii) onset of stable cross-frequency coupling (e.g., theta-gamma PAC), (iii) classifier-detectable state transition in downstream neural populations. We do not claim all neural computation reduces to discrete commits; rather, we claim that the *fastest update of downstream state* is bounded by the time required for multi-module phase alignment. The framework addresses temporal resolution limits, not the full richness of neural dynamics.

Definition (Ontological dimensionality). Throughout this work, "dimensionality" refers to a physical property of the system, not a mathematical measure. Ontological dimensionality is a quantitative measure of effective connectivity: how many independent directions of coordinated variation are accessible to a system given its interaction structure. High-dimensional states correspond to weakly constrained dynamics in which many components vary semi-independently; low-dimensional states correspond to strongly coupled dynamics restricted to a small number of collective modes. Dimensionality in this sense is determined not by component count, but by the rank of the system's interaction structure. Different measurement schemes (participation ratios, manifold dimension, information-geometric metrics) provide projections of this underlying physical property, but do not constitute it [12].

This ontological dimensionality is state-dependent: the same physical system may occupy high- or low-dimensional regimes depending on coupling strength and task demands. Coherence time depends on ontological dimensionality—specifically, how many semi-independent constraints must be simultaneously satisfied for a commit to occur—not on any particular estimator.

1.3 Structure of this work

We first derive the coherence time formula from principles of phase synchronization in coupled oscillator networks (§2.1), then embed it in a unified temporal resolution bound combining quantum, noise, and power constraints (§2.2). Section 2.3 situates our framework explicitly within the quantum and thermodynamic limits of computation, addressing the Landauer bound, Margolus-Levitin quantum speed limit, and energy-time uncertainty relation. Section 3 applies the framework to neural systems, demonstrating quantitative agreement with perceptual binding windows, tachypsychia dissociations, and metabolic scaling. Section 4 discusses the speed-flexibility trade-off, delay embedding as a preservation mechanism, generalization beyond neural systems, and testable predictions.

2 Theory and Methods

2.1 The unified temporal resolution bound

We model biological temporal processing as a sequence of *commits*—thermodynamically irreversible events that register high-dimensional internal state as low-dimensional output. A commit requires: (1) dimensional collapse from distributed state (D_{eff} dimensions) to registered outcome (few dimensions), (2) dissipation of at least $k_{\mathrm{B}}T \ln 2$ per bit (Landauer bound), and (3) creation of a persistent measurement influencing future dynamics.

The minimum time between commits is bounded below by four physical constraints:

$$\boxed{\tau_{\text{eff}} \gtrsim \max\left[\tau_{\text{QSL}},\ \tau_{\text{SNR}},\ \tau_{\text{coh}},\ \tau_{\text{power}}\right]} \tag{1}$$

This is a **lower bound**: the effective commit interval must be at least as long as the slowest constraint. The max operation is appropriate when these constraints act on largely independent subsystems that can be satisfied in parallel (e.g., quantum state evolution, signal detection, and phase coordination proceed concurrently). If constraints were strictly sequential (e.g., signal detection *then* coordination *then* dissipation), a sum would be more appropriate. In biological neural systems, the constraints are largely parallel, and the coherence time dominates by orders of magnitude, so the distinction has minimal practical impact. The four terms are:

Quantum speed limits (τ_{QSL}). The Mandelstam-Tamm and Margolus-Levitin bounds [15, 16] establish

$$\tau_{\text{QSL}} = \max\left[\frac{\hbar}{2\Delta E},\ \frac{\pi\hbar}{2\langle E\rangle}\right]$$

For biological temperatures ($\Delta E \sim k_{\text{B}}T$), $\tau_{\text{QSL}} \sim 10^{-13}$ s.

Signal-to-noise limit (τ_{SNR}). Matched-filter detection in additive white Gaussian noise (two-sided power spectral density N_0, units: power/Hz) over bandwidth B achieves $\text{SNR}(T) = 2P_sT/N_0$ where P_s is signal power and T is integration time [21]. Reaching detection threshold $\rho_{\min} \sim 5$–10 requires

$$\tau_{\text{SNR}} = \frac{\rho_{\min}N_0}{2P_s}$$

Power limit (τ_{power}). Dimensional collapse by $\Delta\mathcal{D}$ at dimensional chemical potential $\lambda_{\mathcal{D}} \sim k_{\text{B}}T_{\text{eff}}$ costs energy $\sim \lambda_{\mathcal{D}}\Delta\mathcal{D}$. At sustained power P, commit rate is bounded by

$$\tau_{\text{power}} = \frac{\lambda_{\mathcal{D}}\Delta\mathcal{D}}{P}$$

Coherence time (τ_{coh}). We model commits as threshold crossings requiring M semi-independent modules to achieve simultaneous phase alignment within tolerance ε. Let r be the mean resultant length (Kuramoto order parameter), and $\kappa(r)$ the von Mises concentration with $r = I_1(\kappa)/I_0(\kappa)$ [23].

Definition of alignment event. We define alignment as **anchored to a reference module**: all modules must fall within $\varepsilon/2$ of a designated reference phase. Formally:

$$A_\varepsilon = \{|\theta_i - \theta_1| < \varepsilon/2 \text{ for } i = 2, \dots, M\}$$

The alignment probability depends on whether the reference is fixed or random:

Case A (anchored/pacemaker reference): If θ_1 is fixed or tightly controlled (e.g., by a pacemaker oscillator, gating signal, or first-crossing module), then

$$P(A_\varepsilon) = \prod_{i=2}^{M} P(|\theta_i - \theta_1| < \varepsilon/2) = p_1(\varepsilon, \kappa)^{M-1}$$

where $p_1(\varepsilon, \kappa) = \int_{-\varepsilon/2}^{\varepsilon/2} \frac{e^{\kappa \cos\theta}}{2\pi I_0(\kappa)} d\theta$ is the single-variable window probability.

Case B (random reference): If θ_1 is itself drawn from the same von Mises distribution, then $P(A_\varepsilon) = \mathbb{E}_{\theta_1}[q(\theta_1)^{M-1}]$ where $q(\theta_1) = P(|\theta - \theta_1| < \varepsilon/2)$. In the small-window limit, rotational symmetry ensures the scaling remains $\propto \varepsilon^{M-1}$, though the constant differs.

Case C (mean-phase reference): Alternatively, alignment can be defined relative to the instantaneous mean phase $\bar{\psi} = \arg(\sum_m e^{i\psi_m})$: all modules must fall within $\varepsilon/2$ of $\bar{\psi}$. This definition is natural for empirical detection (threshold on max phase deviation from mean) and yields the same $(M-1)$ scaling exponent, differing only by an $O(1)$ constant from the anchored case.

In all cases:

$$P(A_\varepsilon) \propto p_1(\varepsilon, \kappa)^{M-1} \tag{2}$$

Small-ε approximation. For small windows, the von Mises density near the mode gives

$p_1(\varepsilon,\kappa) \approx \varepsilon \cdot e^{\kappa}/(2\pi I_0(\kappa))$, yielding:

$$\tau_{\rm coh} \approx \frac{1}{\Delta\omega}\left(\frac{2\pi I_0(\kappa)}{\varepsilon\, e^{\kappa}}\right)^{\alpha(M-1)}$$

This makes the $\varepsilon^{-(M-1)}$ scaling and the "exponential in M" dependence explicit. *Note:* We use the exact $p_1(\varepsilon,\kappa)$ integral for all numerical estimates (e.g., the binding window calculations use $\varepsilon = \pi/2$); the small-ε expansion is presented only to clarify the scaling structure.

The exponent is $(M-1)$, not M. This arises from quotient geometry: the reference module θ_1 serves as anchor (whether fixed or conditioned upon), leaving $M-1$ independent constraints. This is analogous to the classic result that the probability of M uniform phases falling in *some* semicircle is $M/2^{M-1}$, where the factor M accounts for the free choice of arc location; our anchored definition removes this degree of freedom.

First-passage time structure. Assuming independence across modules on the commit timescale, the alignment event A_ε is a rare target set on the M-torus T^M. Under fast mixing relative to the target set size, first-passage times are approximately exponential with mean $\tau \approx 1/(\text{attempt rate} \times P(A_\varepsilon))$—a form of Kac's recurrence theorem [22].

Mixing time interpretation. Let $\tau_{\rm mix}$ denote the mixing time of the joint phase process—the timescale on which phase configurations become approximately independent of their initial state. The effective number of independent "alignment attempts" per unit time is $\sim 1/\tau_{\rm mix}$. We identify $\Delta\omega \approx 1/\tau_{\rm mix}$, yielding:

$$\tau_{\rm coh} = \frac{1}{\Delta\omega}\, p_1(\varepsilon,\kappa(r))^{-(M-1)} \tag{3}$$

For coupled modules (weak inter-module coupling), the effective exponent is reduced:

$$\boxed{\tau_{\rm coh} \approx \frac{1}{\Delta\omega}\, p_1(\varepsilon,\kappa(r))^{-\alpha(M-1)}}, \qquad \alpha \in (0,1], \tag{4}$$

where α captures *effective independence*: $\alpha = 1$ for fully independent modules; $\alpha < 1$ when inter-module coupling induces correlations that reduce the effective number of constraints. One can define an effective constraint dimension $d_{\text{eff}} = \alpha(M-1)$ and estimate it empirically from the participation ratio of the covariance matrix of phase deviations $\delta\theta_i = \theta_i - \bar{\theta}$.

Log-linear form. Taking logarithms makes the exponential-in-M scaling explicit:

$$\log \tau_{\text{coh}} = -\log \Delta\omega - \alpha(M-1)\log p_1(\varepsilon, \kappa(r)).$$

Since $p_1 < 1$, we have $-\log p_1 > 0$, so coherence time grows exponentially with effective constraint dimension $\alpha(M-1)$.

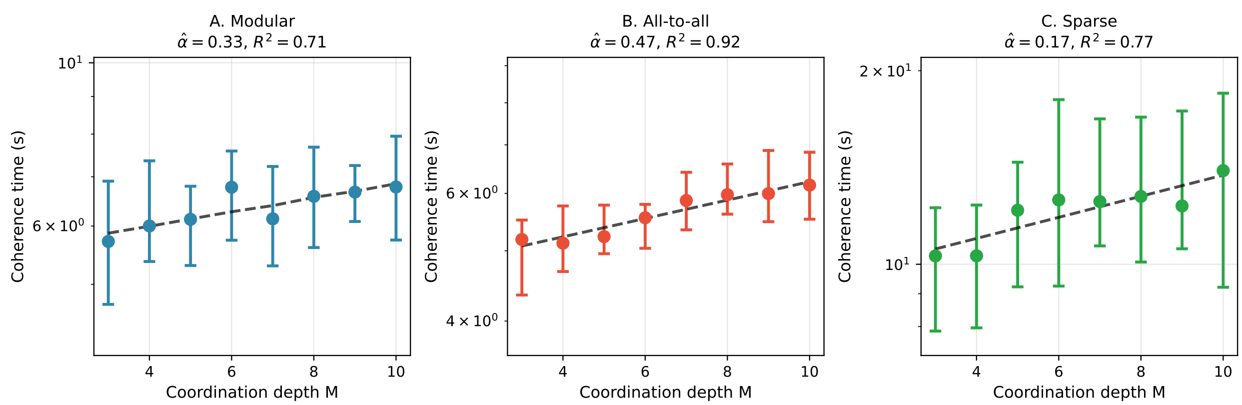

Figure 1: **Validation of M-scaling across network topologies.** Kuramoto oscillator simulations ($N = 100$) confirm exponential scaling of coherence time with coordination depth M. Three connectivity patterns tested: (A) Modular ($\hat{\alpha} = 0.34$, $R^2 = 0.71$), (B) All-to-all ($\hat{\alpha} = 0.47$, $R^2 = 0.93$), (C) Sparse ($\hat{\alpha} = 0.17$, $R^2 = 0.77$). Error bars show interquartile range across 20 trials. Fitted lines use the log-linear model with interpolated $r(M)$. The exponent $\alpha < 1$ reflects inter-module coupling that reduces effective independence.

Key assumptions. Equation 4 relies on:

1. **Stationarity**: The phase distribution is approximately stationary over the commit

timescale.

2. **Fast mixing**: The joint phase process mixes rapidly relative to the target set size, so first-passage times are approximately exponential (memoryless after conditioning on slowly varying parameters like r).

3. **Weak inter-module coupling**: Modules are semi-independent on the commit timescale; strong coupling would synchronize phases and reduce effective M.

4. **Attempt rate interpretation**: $\Delta\omega$ represents the inverse correlation time of phase deviations, not the carrier frequency.

Diagnostic: If these assumptions hold, inter-commit intervals (after conditioning on r) should be approximately exponentially distributed. Deviations indicate strong coupling, metastability, or non-stationarity.

Heavy-tailed waiting times. Empirically observed inter-commit intervals may exhibit heavy tails even when the underlying process is well-described by Eq. 4. This arises naturally when $r(t)$, $\Delta\omega(t)$, or network state drift slowly relative to the commit timescale. The marginal waiting time distribution then becomes a *mixture of exponentials* (a Cox or doubly stochastic process), which can produce power-law-like tails. The key testable prediction is that waiting times *conditional on a narrow bin of* r should remain approximately exponential, even when the unconditional distribution is heavy-tailed.

Physical interpretation: M is the **coordination depth**—the number of semi-independent modules (functional clusters) whose order parameters must jointly exceed a threshold for a commit to register. *Modules are not individual oscillators*: within-module coherence is absorbed into the module's aggregate phase estimate. $\Delta\omega$ is the **phase-exploration rate**: the characteristic frequency at which inter-module phases mix (not the carrier frequency). The tolerance ε is the **full phase window width** (in radians); $\varepsilon = \pi/2$ corresponds to $\pm 45°$ alignment tolerance, typical for gamma-band binding.

Foundational assumption: causal significance of electric fields. This framework presupposes that oscillatory phase relations are *causally significant*—that electric field dynamics influence neural computation beyond what can be captured by action potential timing alone. If neural computation reduced to binary spike events, the substrate would be one-dimensional: spikes either occur or do not. The oscillatory framework we develop here assumes instead that local field potentials and ephaptic coupling [34, 3] carry information and influence computation through their timing structure.

The skeptical position. Several arguments suggest ephaptic effects may be computationally negligible:

1. *Magnitude*: Ephaptic potentials are weak (∼0.1–0.5 mV) compared to synaptic potentials (∼1–10 mV) or action potentials (∼100 mV). The signal is small relative to other influences.

2. *Noise floor*: Neural systems already contend with substantial noise (channel stochasticity, synaptic variability, thermal fluctuations). Ephaptic effects may simply contribute additional noise that gets filtered out.

3. *Redundancy*: Neural circuits have convergent and divergent connectivity. Small timing perturbations in individual neurons should average out across populations.

4. *Synaptic dominance*: Synaptic transmission is the established mechanism of neural communication; ephaptic coupling is at best a second-order correction.

Why these arguments fail at criticality. Each objection assumes a system operating in a robust, noise-tolerant regime. Near criticality [35, 36], the situation inverts:

1. *Magnitude at criticality*: At a critical point, sensitivity diverges. A 0.1 mV shift can advance or delay spike timing by microseconds; in a system poised at a phase transition, microsecond differences determine which neurons cross threshold, which determines downstream cascades. Small causes have large effects—this is what criticality *means*.

2. *Structured vs. random noise*: Ephaptic "noise" is spatially correlated—neighboring neurons experience similar field perturbations. Redundancy and averaging suppress *uncorrelated* noise; correlated perturbations propagate coherently through the network.

3. *Timing vs. connectivity*: Synaptic weights determine *whether* a neuron eventually fires given sufficient input. Fields influence *when* it fires. For oscillatory coordination, timing is the computation. A neuron that fires 2 ms earlier participates in a different phase relationship, potentially a different assembly, a different downstream cascade.

4. *No central clock*: In clocked architectures, timing is imposed externally; perturbations between clock edges are irrelevant. Biological neural systems have no global clock. Timing emerges from dynamics, and fields are part of those dynamics. The question is not whether fields influence spike timing—they demonstrably do—but what mechanism would *prevent* those timing shifts from propagating. In the absence of a clock to enforce timing, there is no such mechanism.

The burden of proof may be inverted: in unclocked systems at criticality, causal significance of field dynamics is the default. Causal *in*significance would require an active suppression mechanism that does not exist. If this argument holds, then oscillators—not spikes—become the fundamental primitive, and the coherence time bounds derived here govern processing rate.

Regional variation in ephaptic coupling. Notably, ephaptic coupling operates differently across brain regions. Cerebellar architecture, with its densely packed parallel fibers and Purkinje cells, supports strong local field effects [37]. Cortical architecture, with its columnar organization and heterogeneous cell types, produces different field geometries. This regional variation in ephaptic coupling strength may explain why cerebellar processing appears exempt from the coordination penalties we attribute to high-M cortical assemblies—the cerebellum may use different coupling mechanisms that bypass the phase-alignment bottleneck.

2.2 The speed-flexibility trade-off

The coherence time reveals opposing constraints on commit rate:

Proposition 1 (Speed-Flexibility Frontier). *At fixed power P, systems face a Pareto frontier in (M, r) space:*

- ***Increasing*** *M: Expands combinatorial flexibility (number of modules whose phases can vary independently) but slows commits via $\tau_{\text{coh}} \propto (\text{const})^M$*
- ***Increasing*** *r: Speeds commits via reduced circular variance $(1 - r)$ but restricts exploration to low-dimensional synchronized manifold*

Increasing coordination depth M expands the dimensionality of commit-registered outcomes but requires exponentially longer alignment time. Increasing coherence r speeds commits but forces trajectories onto low-dimensional attractors, reducing the set of accessible commit outcomes.

2.3 Quantum and thermodynamic limits of computation

A natural question is how this coordination-based framework relates to fundamental physical limits on computation. We now explicitly address the quantum and thermodynamic bounds that form the basis of information transmission in living systems.

2.3.1 The Landauer bound

Landauer's principle [13] establishes that any logically irreversible operation—specifically, the erasure or overwriting of one bit of information—dissipates at least $k_{\text{B}}T \ln 2$ of energy as heat. This bound arises from the connection between logical irreversibility and thermodynamic entropy: erasing a bit reduces the system's information-theoretic entropy, which must be compensated by entropy increase in the environment [14].

The minimum energy per bit at biological temperature ($T \approx 310$ K) is:

$$E_{\text{Landauer}} = k_{\text{B}} T \ln 2 \approx 3 \times 10^{-21} \text{ J/bit}$$

This sets an absolute floor on energy dissipation per irreversible operation. However, the Landauer bound constrains energy per operation, not time per operation. A system could in principle perform operations arbitrarily fast if sufficient power were available—until quantum speed limits intervene.

2.3.2 The Margolus-Levitin bound and quantum speed limits

The Margolus-Levitin theorem [16] establishes that the minimum time to transition between orthogonal quantum states is:

$$\tau_{\text{ML}} = \frac{\pi \hbar}{2\langle E \rangle}$$

where $\langle E \rangle$ is the average energy above the ground state. This bound, together with the Mandelstam-Tamm relation [15] ($\tau_{\text{MT}} = \hbar/(2\Delta E)$ where ΔE is energy uncertainty), constitutes the quantum speed limit on state evolution.

These bounds originate from the energy-time uncertainty relation $\Delta E \cdot \Delta t \gtrsim \hbar/2$, which is fundamental to understanding information transmission in physical systems [17, 18, 19]. As Bormashenko [14] notes, synthesis of the Landauer bound with the Margolus-Levitin limit yields a minimal computation time scaling as $\tau_{\min} \sim h/k_{\text{B}}T$—decreasing temperature reduces energy consumption per operation but simultaneously slows computation.

At biological temperatures with $\langle E \rangle \sim k_{\text{B}} T \approx 4 \times 10^{-21}$ J:

$$\tau_{\text{QSL}} \sim \frac{\pi \times 10^{-34}}{2 \times 4 \times 10^{-21}} \sim 10^{-13} \text{ s}$$

This is fourteen orders of magnitude faster than the millisecond timescales of neural processing.

2.3.3 Why coherence time dominates in biological systems

The central claim of this paper is that in biological neural systems, the quantum speed limit and Landauer bound are typically **non-binding constraints**. The dominant bottleneck is coherence time—the time required for distributed oscillatory assemblies to achieve sufficient phase coordination for a commit to occur.

Consider the hierarchy of timescales for cortical visual processing:

$$\tau_{\mathrm{QSL}} \sim 10^{-13}\ \mathrm{s} \quad \text{(quantum speed limit)} \tag{5}$$

$$\tau_{\mathrm{SNR}} \sim 10^{-3}\ \mathrm{s} \quad \text{(signal detection)} \tag{6}$$

$$\tau_{\mathrm{coh}} \sim 10^{-2}\ \mathrm{s} \quad \text{(coordination time)} \tag{7}$$

(Note: The Landauer bound constrains *energy* per bit, not time; we omit it from this hierarchy since the time-limiting step is coordination, not dissipation.)

The coherence time exceeds quantum limits by eleven orders of magnitude. This enormous gap arises because neural computation is not limited by how fast individual elements can switch states, but by how long it takes for *distributed assemblies of semi-independent modules* to achieve coordinated phase alignment.

Mathematically, this reflects the exponential scaling of coherence time with coordination depth:

$$\tau_{\mathrm{coh}} \propto p_1^{-\alpha(M-1)}$$

For $M = 8$ modules with $p_1 \approx 0.7$ and $\alpha = 0.7$, the exponential factor is $(0.7)^{-4.9} \approx 6$. Even modest coordination requirements ($M \sim 5$–15) produce millisecond-scale coherence times from microsecond-scale phase dynamics.

This analysis clarifies the relationship between our framework and fundamental physics: quantum and thermodynamic limits set absolute bounds on computation, but biological systems operate in a regime where multi-module coordination—not quantum switching speed or Landauer dissipation—determines processing rate.

2.4 Parameter estimation protocols

Coordination depth (M). M is the count of semi-independent modules whose order parameters must jointly exceed a threshold to register a commit. **Operational estimation:**

1. Cluster channels/units by band-limited coherence or Granger causality into modules.
2. Identify the minimal set of modules whose joint activity predicts behavioral commits.
3. Typical values: occipital visual tasks yield $M \sim 6$–10; cross-modal binding $M \sim 10$–15.

Kuramoto coherence (r). Extract instantaneous phase $\theta_j(t)$ from bandpass-filtered LFP/EEG via Hilbert transform. For the coherence time model, we distinguish:

- **Within-module coherence:** $r_{\text{intra},m}(t) = \left|\frac{1}{n_m}\sum_{j\in m} e^{i\theta_j(t)}\right|$ for module m with n_m channels.
- **Inter-module coherence:** $r_{\text{inter}}(t) = \left|\frac{1}{M}\sum_{m=1}^{M} e^{i\psi_m(t)}\right|$ where ψ_m is module m's aggregate phase.

Commits require high r_{intra} (modules are internally coherent) *and* rare alignment of inter-module phases (low baseline r_{inter}, with commits occurring when r_{inter} transiently exceeds threshold). The r and κ in Eq. 4 refer to the *inter-module* distribution. Typical inter-module values: spontaneous cortex ~ 0.2–0.4; attention ~ 0.5–0.7 [4].

Interpretation of r in Eq. 4. The parameter r can be interpreted in two ways: (1) as a slowly varying *state variable* $r(t)$, in which case Eq. 4 gives an instantaneous hazard rate $\lambda(t) = \Delta\omega\, p_1^{\alpha(M-1)}$, and the waiting time distribution becomes a hazard integral over the trajectory of $r(t)$; or (2) as a *conditional* parameter representing coherence during the pre-commit interval, in which case Eq. 4 gives the mean waiting time conditional on that coherence level. Interpretation (1) naturally explains heavy-tailed waiting time distributions when $r(t)$ drifts slowly.

Phase alignment window (ε). From spike train or LFP cross-correlograms, measure half-width at half-maximum (HWHM, units: seconds). The **full** phase window is $\varepsilon = 2\,\omega_0\,\mathrm{HWHM}$ (radians).

Phase-exploration rate ($\Delta\omega$). Estimate from instantaneous frequency spread or phase diffusion. Typical values: $\Delta\omega \sim 2\pi \times (5\text{–}20)$ rad/s for cortical gamma/alpha bands.

3 Results

3.1 Visual perceptual binding windows

Human visual perception exhibits temporal integration windows of 30–50 ms (flicker fusion at 20–30 Hz). We apply Eq. 1 with neural parameters.

Parameter provenance: We take $M = 8$ occipital modules (a plausible count of functional submodules in visual binding, including V1, V2, V4, and posterior parietal subdivisions; can be estimated via community detection on coherence networks). Coherence $r = 0.7$–0.8 reflects attentional engagement [4]. Full tolerance $\varepsilon = \pi/2$ rad ($\pm 45^\circ$) corresponds to the phase windows observed in gamma-band cross-correlograms during visual binding; note that $\pm 90^\circ$ tolerance would be too loose for functional synchrony. Phase-exploration rate $\Delta\omega = 2\pi \times 20$ rad/s is estimated from the decorrelation time of relative phases in alpha/beta bands (not the carrier frequency). Effective independence $\alpha = 0.7$ reflects modular cortical topology; it can be fit empirically from the scaling of inter-commit times with M.

First compute p_1: at $r = 0.7$, $\kappa \approx 2.0$, giving $p_1(\pi/2, 2.0) \approx 0.68$ (numerical integration of Eq. 2). Using Eq. 4 with $\alpha = 0.7$:

$$\tau_{\mathrm{coh}} \approx \frac{1}{125.66}\,(0.68)^{-0.7\times 7} = \frac{1}{125.66}\,(0.68)^{-4.9} \approx 54 \text{ ms}.$$

With higher coherence $r = 0.8$ ($\kappa \approx 2.9$, $p_1(\pi/2, 2.9) \approx 0.78$):

$$\tau_{\mathrm{coh}} \approx \frac{1}{125.66}\,(0.78)^{-4.9} \approx 27 \text{ ms.}$$

Computing all terms in Eq. 1:

$$\tau_{\mathrm{QSL}} \sim 10^{-13}\,\mathrm{s} \quad \text{(negligible—quantum limit non-binding)} \tag{8}$$

$$\tau_{\mathrm{SNR}} \sim 5\text{–}10 \text{ ms} \quad \text{(task-dependent)} \tag{9}$$

$$\tau_{\mathrm{power}} \ll 1 \text{ ms} \quad \text{(per commit)} \tag{10}$$

$$\tau_{\mathrm{coh}} \approx 27\text{–}54 \text{ ms} \quad (r = 0.7\text{–}0.8,\ M = 8) \tag{11}$$

The effective commit time is $\tau_{\mathrm{eff}} \gtrsim \max[\text{QSL, SNR, power, coh}] \approx 27\text{–}54$ ms, matching human binding windows (30–50 ms). **Coherence time dominates**; quantum and Landauer bounds are not the limiting factors.

Sensitivity analysis. Table 1 shows how τ_{coh} varies across plausible parameter ranges. The 20–80 ms binding window is robust to parameter uncertainty. All p_1 values computed via numerical integration of Eq. 2; κ obtained from r via numerical inversion of $r = I_1(\kappa)/I_0(\kappa)$.

Sanity check: For any $\kappa \geq 0$, $p_1(\pi, \kappa) \geq 0.5$ with strict inequality for $\kappa > 0$, since the window $\varepsilon = \pi$ covers half the circle centered on the mode of a unimodal distribution.

Table 1: Sensitivity of coherence time to parameter variations. Base case: $M = 8$, $r = 0.75$, $\varepsilon = \pi/2$, $\Delta\omega = 2\pi \times 20$ rad/s, $\alpha = 0.7$.

Parameter varied	M	r	ε	$\Delta\omega$ (rad/s)	τ_{coh} (ms)
Base case	8	0.75	$\pi/2$	$2\pi \times 20$	39
Lower M	6	0.75	$\pi/2$	$2\pi \times 20$	25
Higher M	10	0.75	$\pi/2$	$2\pi \times 20$	61
Lower r	8	0.65	$\pi/2$	$2\pi \times 20$	75
Higher r	8	0.85	$\pi/2$	$2\pi \times 20$	19
Narrower ε	8	0.75	$\pi/3$	$2\pi \times 20$	105
Faster $\Delta\omega$	8	0.75	$\pi/2$	$2\pi \times 30$	26

The exponential dependence on M and strong dependence on r mean that modest changes in attention state (affecting r) or task complexity (affecting M) can shift binding windows by factors of 2–4, consistent with empirical observations of attention-dependent temporal resolution [25].

3.2 Tachypsychia: a dual-loop hypothesis

During acute stress or falls, subjects report subjective time slowing while objective reaction times remain unchanged [24]. We propose a **dual-commit architecture** as one explanation for this dissociation (alternative accounts exist, including attentional sampling rate changes and memory density effects):

Perceptual loop (cortical): High-M ($\sim$ 10–15 modules) coherent field dynamics. Commits sparse (5–20 Hz). Arousal increases coherence r and thus information rate per commit interval.

Motor loop (cerebellar/basal ganglia): Low-M ($\sim$ 3–5 modules) primitives. Motor latencies of 50–150 ms reflect simpler coordination requirements (lower M, higher α) than multi-area perceptual binding. Arousal modulates decision threshold, preserving reaction time.

Measurable proxies: Perceptual loop commit rate could be indexed by temporal-order judgment thresholds or EEG phase coherence in theta/alpha bands. Motor loop could be indexed by simple RT distributions or EMG onset latencies. The hypothesis predicts these measures should be *statistically dissociable* under arousal manipulation, with temporal-order thresholds changing while simple RT remains stable.

3.3 Alpha oscillations and temporal acuity

Alpha oscillations (8–12 Hz) in visual cortex correlate with temporal acuity: individuals with faster alpha perceive time faster [25, 26]. One interpretation is that alpha acts as a gating rhythm that modulates commit opportunity windows, effectively setting the attempt rate $\Delta\omega$ or the sampling schedule for phase alignment.

Prediction: If alpha frequency modulates commit timing, then entraining alpha via transcranial alternating current stimulation (tACS) should shift perceptual binding windows approximately linearly with frequency change. This prediction distinguishes the gating interpretation from alternatives where alpha reflects post-commit memory encoding.

3.4 Metabolic scaling across species

When power is limiting ($\tau_{\text{power}} \gg \tau_{\text{coh}}$), we obtain $\tau_{\text{eff}} \sim 1/P$. Critical flicker fusion frequency should scale with mass-specific metabolic rate.

Healy et al. [27] measured critical flicker fusion frequency across diverse taxa: flies ~ 240 Hz, humans ~ 60 Hz, leatherback turtles ~ 15 Hz. Their analysis found that mass-specific metabolic rate explains substantial variance in CFF across species spanning three orders of magnitude in body mass, consistent with the predicted power-limited regime for small, high-metabolism animals.

4 Discussion

4.1 Biological examples of the speed-flexibility trade-off

Mind-wandering and creative insight (high M, low r). Default-mode network activity exhibits low inter-regional coherence ($r \sim 0.3$) and high coordination depth ($M \sim 12$–20). Commits are rare (~ 0.1–1 Hz), producing subjective "slow thinking" but enabling novel associations.

Focused attention (moderate M, moderate r). Attention increases coherence ($r : 0.3 \to 0.6$) and narrows coordination to task-relevant modules ($M : 15 \to 8$) [4]. This produces faster commits (10–30 Hz) in a restricted subspace.

Automaticity and skill learning (low M, high r). Overlearned motor sequences compress to low-dimensional primitives ($M \sim 3$–5) with tight coherence ($r \sim 0.8$). Commits are rapid (50–150 ms) but inflexible.

4.2 Temporal embedding as a pre-commit representation mechanism (hypothesis)

Coherence time tells us *how long* pre-commit dynamics must be maintained before alignment occurs. A separate question is *how* high-dimensional structure can remain accessible during that interval. We propose—as a hypothesis for future investigation—that biological systems may implement **Takens delay embedding** [28].

Takens' theorem. For a deterministic system with state dimension d, time-delayed copies of a scalar observable—$x(t), x(t-\tau), x(t-2\tau), \ldots$—can reconstruct the complete state-space topology. Crucially, $2d+1$ delay coordinates generically suffice.

A candidate biological implementation: cerebellar parallel fibers. Parallel fibers—unmyelinated granule cell axons—conduct at 0.2–0.5 m/s with systematically varying path lengths, creating conduction delays of 5–15 ms [29, 30, 31]. These delays match the optimal embedding parameter $\tau^* = 1/(4f)$ for motor dynamics at 15–50 Hz.

We hypothesize that the cerebellum may provide one biological implementation of delay-embedded state preservation during pre-commit intervals. This would enable rapid low-dimensional readout (motor commands) from high-dimensional state without requiring simultaneous global coordination. **This remains conjecture**; direct evidence would require demonstrating that cerebellar computations exploit delay structure for state reconstruction.

Robustness to noise. We use Takens embedding as a *design principle* rather than a literal guarantee: delay-line architectures improve state inference under partial observability, even when exact reconstruction is impossible.

Connection to coherence time. Coherence time determines the maximum window over which temporal embeddings remain usable. If coherence decays faster than the embedding window, reconstruction fails and high-dimensional structure is lost before a commit can occur.

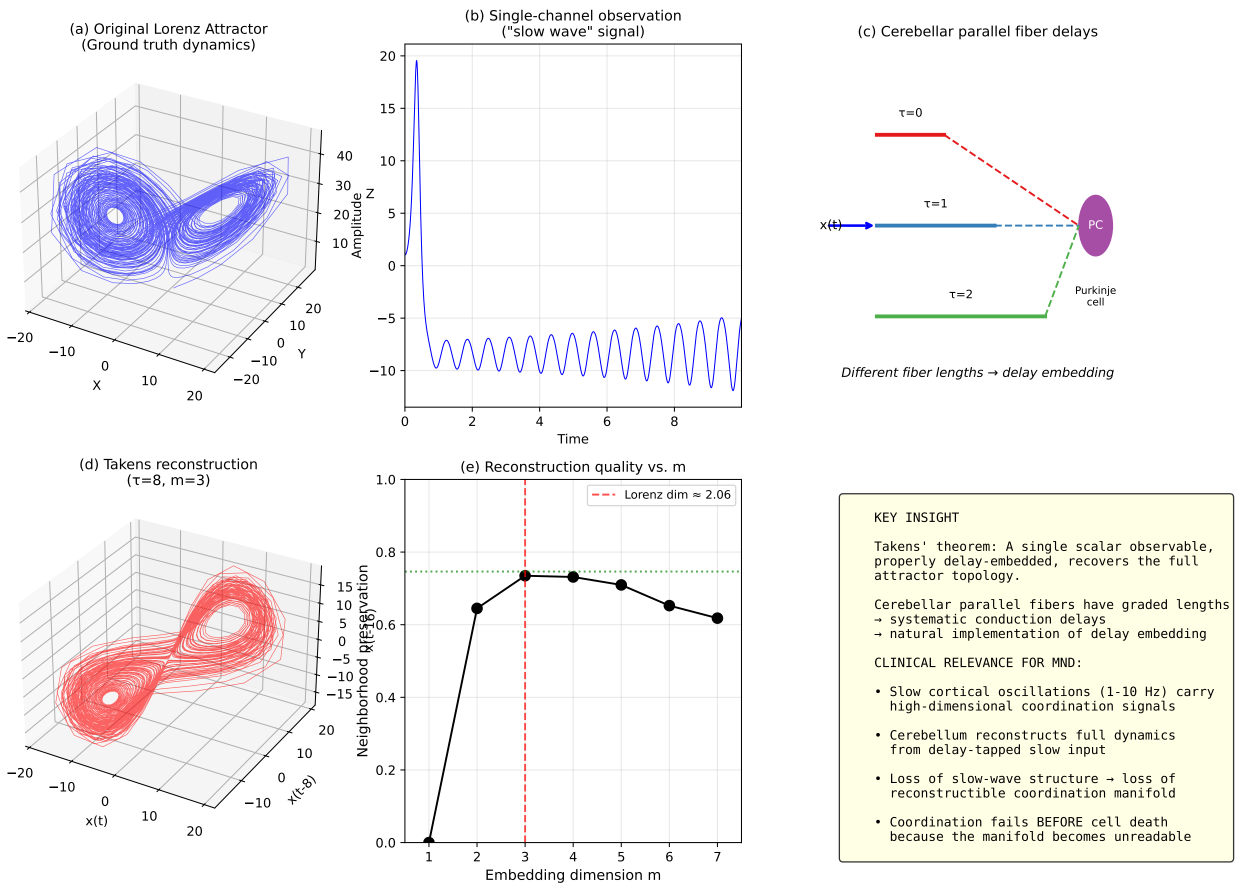

Figure 2: **Delay embedding reconstructs high-dimensional dynamics from scalar observation.** Simulation demonstrating Takens' theorem applied to cerebellar architecture. (a-b) Lorenz attractor (ground truth, $d \approx 2.06$) and its 1D projection (what a slow cortical oscillation might carry). (c-d) Delay embedding with $\tau = 8$, $m = 3$ reconstructs attractor topology from the scalar signal alone. (e) Neighborhood preservation quantifies reconstruction quality across embedding dimensions. (f) Theoretical implications: parallel fibers with graded conduction delays implement delay embedding, enabling reconstruction of high-dimensional coordination signals from low-frequency input.

4.3 Generalization beyond neural systems

While we emphasize neural applications given their empirical accessibility, the coherence time framework applies to any biological system whose function requires coordinated phase relations and whose outputs are sparse commitment events.

Scope statement. Neural systems are the best-instrumented instance of a broader class: living, self-maintaining oscillator assemblies that perform irreversible state registration. We do not treat collectives (e.g., societies) as intelligent systems in this sense; they are coordination networks of intelligent organisms and introduce additional channels not modeled here.

Chromosomal coordination. Chromosomal systems provide a non-neural example of the same architecture. High-dimensional continuous dynamics (polymer conformations, electrostatic interactions, topological constraints, phase separation) produce function via sparse low-dimensional commits (origin firing, checkpoint decisions, segregation outcomes).

Anaphase onset is naturally modeled as a commit event: a global irreversible transition triggered only after multiple semi-independent units satisfy a coherence-like constraint. The waiting time should scale steeply with effective coordination depth, consistent with first-passage into a small target set. The relevant "order parameter" is any coordination metric determining whether a global event can proceed—spatial colocalization, topological state, mechanical alignment—not necessarily Kuramoto r.

Circadian networks. Circadian gene networks exhibit $M \sim 3$–5 core feedback loops, $r \sim 0.8$ (tight coupling), and $\varepsilon \sim 0.5$ rad (loose phase tolerance). These parameters yield coordination times much shorter than the 24-hour period, implying coordination is *not* the bottleneck. The 24-hour period arises from molecular lags. The framework correctly identifies the dominant regime: circadian clocks are biochemical-delay-limited, not coordination-limited.

4.4 Testable predictions

Alpha entrainment. tACS at alpha frequency should shift perceptual binding windows linearly with frequency change. 20% frequency increase predicts 20% threshold reduction.

Arousal effects on dual-loop dissociation. During simultaneous simple reaction time + temporal-order judgment under acute stress:

- Temporal-order discrimination threshold changes
- Simple reaction time unchanged or slightly faster
- Statistical independence between measures

Coherence measures and consciousness. If phenomenal experience reflects pre-commit coherent dynamics, conscious states should correlate more strongly with phase-locking index and theta-gamma coupling than with spike count or mean firing rate.

4.5 Relationship to existing frameworks

Quantum-like cognition. Recent work has explored connections between oscillatory neural networks and quantum-like cognitive phenomena [20]. Our framework provides a classical high-dimensional explanation for why such quantum-like signatures emerge: high-dimensional dynamics with limited measurement bandwidth produce phenomenology parallel to quantum systems—measurement-like collapse at commits, superposition-like coexistence of states pre-commit, uncertainty-like timing inaccessibility.

Integrated Information Theory [32]: We provide thermodynamic/temporal foundations. Integration costs energy and takes time (exponentially with coordination depth).

Free Energy Principle [33]: We add temporal structure. Prediction updates occur at commit events with rate limited by Eq. 1.

4.6 Frequency and dimensionality: a disagreement with current theory

A prominent view in systems neuroscience holds that low-frequency oscillations (delta, theta, alpha) support low-dimensional, "simple" dynamics, while high-frequency oscillations (gamma) support high-dimensional, "complex" computations [1, 2]. Miller and colleagues argue that slow oscillations provide a low-dimensional "scaffold" or "broadcast" signal, while fast gamma carries detailed content in a higher-dimensional space. Recent work extends this framework to cytoelectric coupling, proposing that electric fields organize neural activity via ephaptic effects down to the cytoskeletal level [3].

We disagree. The coherence time framework predicts the opposite relationship between frequency and dimensionality:

The physics argument. Consider an oscillatory system that must achieve phase alignment to register a commit. The number of alignment "attempts" per oscillation cycle is roughly $f/\Delta\omega$, where f is the carrier frequency and $\Delta\omega$ is the phase-exploration rate. For alignment probability p_1:

- **High frequency**: Few attempts per cycle. To achieve reliable alignment within one period, the system requires high p_1, which requires high coherence r. But high r confines dynamics to a low-dimensional synchronized manifold. *Fast oscillations force low dimensionality.*

- **Low frequency**: Many attempts per cycle. The system can tolerate low p_1 and thus low r. Phases can explore more of configuration space while still achieving alignment before the next cycle. *Slow oscillations permit high dimensionality.*

This is not a correlation but a constraint: maintaining high-dimensional coherent dynamics at high frequency is physically harder because there is less time for the rare alignment events that preserve high-M coordination.

Reinterpreting the empirical evidence. The observation that gamma activity correlates with "detailed" processing and alpha/theta with "global" states is consistent with our framework but admits a different interpretation. High-frequency gamma may reflect *local*, low-M modules committing rapidly—not high-dimensional integration but many parallel low-dimensional commits. Meanwhile, low-frequency oscillations may coordinate high-M assemblies that *require* slow timescales to achieve alignment.

The observability problem. Claims about neural dimensionality face a fundamental measurement limitation: electrodes are sparse, noisy, low-dimensional sensors. One can only estimate dimensionality up to the rank of the observation matrix. If slow oscillations genuinely support high-dimensional dynamics (as our framework predicts), this would not necessarily be visible through electrode recordings—the measurement apparatus itself constrains what can be observed.

Consider an analogy: the gravitational field at a point is a 3-dimensional vector, yet it integrates information about the distribution of all mass in the universe. The low-dimensional measurement (a vector) does not imply low-dimensional underlying structure. Similarly, a slow oscillation measured as a scalar or low-rank signal may nonetheless coordinate high-dimensional dynamics that project onto that measurement. The claim that "gamma is high-dimensional" may reduce to "gamma is the highest-dimensional signal we can resolve with our electrodes"—a statement about the measurement, not the system.

This reflects a general methodological point: physics uses low-dimensional mathematics because it is computable and because our instruments measure low-dimensional projections. The predictive success of low-dimensional models demonstrates that low-dimensional descriptions are *useful*, not that nature is low-dimensional. The same applies to neuroscience: the success of low-rank decompositions (PCA, factor analysis, demixed PCA) in capturing behaviorally relevant variance does not establish that neural dynamics are intrinsically low-dimensional—it establishes that low-dimensional projections correlate with the behavioral variables we measure. These are epistemological claims about our models, not ontological claims about the system.

Testable predictions distinguishing the two views.

1. **Dimensionality-frequency relationship**: If Miller et al. are correct, pharmacologically or optogenetically increasing gamma power should increase state-space dimensionality (e.g., participation ratio of neural covariance). If our framework is correct, increasing gamma power while holding coordination depth M constant should *decrease* dimensionality (forcing tighter coupling to maintain coordination at higher frequency).

2. **Information content per cycle**: Miller et al. predict more bits per gamma cycle than per alpha cycle. We predict the opposite: more bits per alpha cycle (because high-M commits are possible), fewer bits per gamma cycle (because only low-M commits can keep pace).

3. **Cross-frequency coupling direction**: If slow oscillations scaffold fast ones (Miller), gamma amplitude should be phase-locked to theta/alpha with high fidelity. If fast oscillations are dimensionally constrained (our view), gamma bursts should occur when local modules achieve rapid low-M commits, potentially *out of phase* with slow oscillations that coordinate broader assemblies.

We note that this disagreement is empirically tractable. The coherence time framework makes quantitative predictions about the relationship between oscillation frequency, coupling strength, and state-space dimensionality that can be tested with existing recording and analysis methods.

4.7 Implications for artificial systems

The coherence time framework identifies a fundamental design constraint for any distributed computing system that lacks a global clock.

Clocked architectures. Conventional digital computers (CPUs, GPUs, current AI accelerators) enforce synchronization via a global clock signal. Every operation is gated by a central timing reference; state updates occur in lockstep across all components. This architecture sidesteps the coherence time penalty entirely—there is no waiting for spontaneous phase alignment because alignment is *imposed* by the clock.

We hypothesize that this comes at a cost beyond power and wire delays: **clocked systems may be constrained in ontological dimensionality**. The clock imposes a global update constraint that can suppress the kinds of metastable, spontaneously coordinated dynamics we model here. Even massively parallel operations (matrix multiplications across thousands of cores) proceed via synchronized state updates—the system can *represent* high-dimensional states in memory, but may not *instantiate* the autonomous, semi-independent module dynamics that characterize biological oscillator assemblies. This is a hypothesis about dynamical structure, not a theorem; intermediate architectures (asynchronous digital, neuromorphic with local clocks) occupy a spectrum between the extremes we describe.

Unclocked architectures. Biological neural networks, and potentially neuromorphic hardware designed on similar principles, operate without global synchronization. Each module evolves on its own trajectory; the phase space is the *actual* dynamical space of the system, not a represented abstraction. The dimensionality is ontologically realized in the substrate.

Such systems pay the coherence time penalty derived in this paper: coordination across M modules requires waiting for rare alignment events, with time scaling exponentially in M. But they gain access to genuinely high-dimensional dynamics—the very property that prior work identified as constitutive of intelligence [11].

A design hypothesis. This suggests a potential trade-off for artificial systems:

- **Clocked**: Fast, deterministic, scalable in parallelism—but potentially constrained in dynamical autonomy. The substrate may not support the spontaneous coordination dynamics modeled here.

- **Unclocked**: Capable of genuinely autonomous, high-dimensional dynamics—but subject to coordination bottlenecks that scale exponentially with integration depth.

If high-dimensional coherent dynamics prove to be not merely *correlated with* but *constitutive of* certain cognitive capacities, then clocked architectures may face constraints that unclocked ones do not. This remains speculative; the empirical question is whether the coordination-time framework applies to engineered systems and, if so, whether it identifies real performance boundaries.

This framing may inform the design of neuromorphic systems, spiking neural networks, and other non-von Neumann architectures. The coherence time formula (Eq. 4) provides quantitative guidance: to achieve update rates comparable to biological cognition, such systems must either accept modest coordination depth M, engineer high baseline coherence r, or develop hierarchical architectures that localize coordination requirements.

The cerebellum as biological precedent. The mammalian cerebellum exemplifies this hierarchical solution. Operating with low coordination depth ($M \sim 3$–5 modules), tight coupling ($r \sim 0.8$), and stereotyped primitives, the cerebellum achieves commit times of 10–20 ms—fast enough to override slow, high-dimensional cortical dynamics when rapid action is required. Skilled movement, postural correction, and overlearned sequences route through this low-M pathway precisely because cortical coordination cannot keep pace.

Notably, cerebellar processing is largely *unconscious*. We do not experience the moment-to-moment adjustments during a golf swing or the postural corrections that keep us upright. If phenomenal experience is associated with high-dimensional pre-commit dynamics (as suggested in §4.4), then low-dimensional cerebellar override would be expected to fall outside awareness. The architecture that solves the speed problem—collapsing to low M for fast commits—simultaneously removes processing from the domain of conscious experience. This is not a bug but a feature: rapid action requires precisely the dimensional compression that excludes phenomenology.

5 Conclusion

We have established that coherence time—the time required for distributed oscillatory assemblies to achieve sufficient phase alignment for irreversible state registration—bounds the rate of neural processes associated with cognition. The lower bound

$$\tau_{\text{eff}} \gtrsim \max\left[\tau_{\text{QSL}},\, \tau_{\text{SNR}},\, \tau_{\text{coh}},\, \tau_{\text{power}}\right]$$

with coherence time

$$\tau_{\text{coh}} = \frac{1}{\Delta\omega}\, p_1(\varepsilon, \kappa(r))^{-\alpha(M-1)}$$

unifies quantum, noise, and coordination limits to explain temporal resolution in biological processing systems.

We have situated this framework explicitly within the hierarchy of physical limits on computation: the Landauer bound constrains energy per irreversible operation; the Margolus-Levitin and Mandelstam-Tamm bounds constrain the speed of quantum state evolution; the energy-time uncertainty relation underlies both. In biological neural systems, these quantum and thermodynamic bounds are typically non-binding. The dominant constraint is the time required for multi-module coordination—a classical phenomenon arising from rare-event statistics on product manifolds.

Key findings:

1. **Speed-flexibility trade-off**: Increasing M exponentially slows commits but expands combinatorial flexibility. Increasing r speeds commits but restricts dynamics to low-dimensional manifolds.

2. **Quantum limits non-binding**: Neural coherence times exceed quantum speed limits by eleven orders of magnitude. Coordination, not quantum switching, limits processing rate.

3. **Generality**: While neural systems provide the empirical testbed, the framework ap-

plies to any biological oscillator assembly with coordinated phase relations and sparse commits.

4. **Quantitative predictions**: Visual binding windows (30–50 ms), metabolic scaling of temporal acuity, alpha entrainment linearity, dual-task dissociations.

The central thesis: biological systems trade speed for flexibility by moving on the coherence-dimension frontier, operating in a regime where quantum and thermodynamic limits permit far faster processing than coordination constraints allow.

Data accessibility

This is a theoretical framework paper. Kuramoto simulation code and parameter estimation workflows are available at `https://github.com/todd866/coherence-time`.

Competing interests

The author declares no competing interests.

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